

Khan Mohd Aasim Maqsood

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EXPERIENCE

Google Summer of Code - IOOS

Open Source Software Developer

Jun 2025 – Aug 2025

github.com/phytoclass

- Contributed to **Phytoclass**, a production research platform for phytoplankton analysis, delivering **15+ pull requests** and resolving **10+ defects** across the R package and Shiny GUI.
- Eliminated **cross-session I/O collisions** during **concurrent analysis** by implementing **session-based file isolation**, creating separate workspaces for each session and validating isolation across **25+ sessions**.
- Hardened the **analysis pipeline** with matrix validation and boundary checks, preventing malformed inputs and automating **algorithm selection** based on dataset characteristics.
- Integrated **reproducibility and CI workflows** using **testthat** and GitHub Actions, automatically regression-testing analysis outputs across random seeds and documenting custom-data workflows for maintainability.

PROJECTS

EVUA - Legacy Code Modernization Platform

Python, FastAPI, React, MySQL

Jul 2025 – Apr 2026

github.com/aasimkhan02/EVUA

- Automated **legacy code modernization** for AngularJS and PHP applications by engineering a Python/FastAPI migration engine with **AST-based transformations**, validated against **200+ source-file test cases**.
- Engineered **40+ AST transformation rules** for automated refactoring, with **LLM-assisted fallbacks** to handle unsupported code patterns and edge cases across legacy codebases.
- Achieved **85% transformation accuracy** on real-world GitHub repositories by validating automated migrations across diverse legacy code patterns and verifying transformed code for correctness.

ThreatBox - AI-Powered Malware Analysis Platform

Go, Python, FastAPI, PostgreSQL, Windows Sandbox, React

Aug 2026 – Present

github.com/aasimkhan02/ThreatBox

- Developed an automated **malware analysis platform** with Go-based job orchestration, **Python/FastAPI services**, PostgreSQL-backed analysis pipelines, and isolated **Windows Sandbox** execution for safely analyzing suspicious files.
- Collected and analyzed **behavioral telemetry** across process, file, network, and system activity, with **IOC extraction** and MITRE ATT&CK mapping for detected behaviors.
- Integrated **threat scoring, AI-generated reports, WebSocket event streaming, and React dashboards** to deliver real-time analysis results and behavioral investigation.

TECHNICAL SKILLS

Languages: Go, Python, SQL, JavaScript, C, R

Backend: FastAPI, REST APIs

Frontend: React, HTML, CSS

Databases: PostgreSQL, MySQL, Redis

Cloud & DevOps: AWS, Docker, Git, GitHub Actions

EDUCATION

Anjuman-i-islam Kalsekar Technical Campus

Bachelor of Computer Engineering, Honors in Cyber Security

Jul 2022 – Jun 2026

CGPA: 8.1 / 10

CERTIFICATIONS & ACHIEVEMENTS

- Received the **Best Presenter** award at a college-level **Smart India Hackathon (SIH)** competition.
- Solved 600+ problems on LeetCode** covering data structures and algorithms.
- Oracle Cloud Infrastructure Developer (Professional)**
- System Design** – Pearson – Coursera